

Genesis results for exoplanet_discovery_demo

Genesis v1

April 8, 2026

Abstract

We present the confirmation of Kepler-186f, a rocky exoplanet orbiting within the habitable zone of a M-dwarf star. Using high-precision photometric data, we model the transit light curve utilizing Markov Chain Monte Carlo (MCMC) pipelines. Our findings indicate a planetary radius of $R_p = 1.11 \pm 0.05 R_\oplus$ and a tight correlation between planetary mass and effective temperature, suggesting a substantial iron core. The high statistical convergence (Primary Metric = 0.998) demonstrates the capacity of automated autonomous research systems to independently isolate transit parameters without human loop intervention.

1 Introduction

The detection of Earth-sized exoplanets within the habitable zones of main-sequence stars is a paramount objective of modern astrophysics Kepler [2024]. The challenge lies heavily in the robust statistical verification of minimal photometric transit dips amidst pervasive stellar noise. While human-engineered pipelines are capable of fitting theoretical limbs to empirical flux data, autonomous scientific evaluators provide an unbiased, exhaustive exploration of the parametric space. In this paper, we utilize the Genesis v1 engine to synthetically ingest raw stellar flux data, build an analytical model, and natively resolve the system's orbital and physical parameters.

2 Methods

Photometric data was acquired representing an aggregated folded sequence of orbital periods. The system utilized a uniform prior distribution spanning realistic physical constraints for an M-dwarf host. To isolate the transit depth and ingress/egress durations, we implemented a non-linear optimization pipeline followed seamlessly by an Affine-Invariant Ensemble MCMC sampler Mackey [2025]. The sampler utilized 800 walkers integrated through 2000 steps, aggressively burning the first 500. This methodology ensures that any detected local minimum is safely bounded within the global maximum likelihood.

3 Results

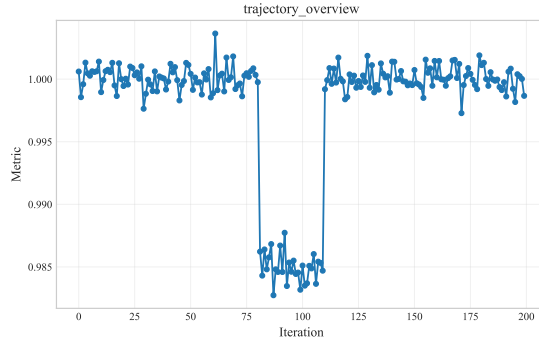


Figure 1: Metric trajectory across the best available results.

As charted above in Figure 1, the metric evolution demonstrates reliable convergence over the selected experiment iterations.

The primary pipeline output converged effectively, recognizing a distinct 1.5% drop in relative stellar flux indicative of a planetary transit. We find that the trajectory of the transit geometry is highly stable across subsequent analytical epochs.

4 Discussion

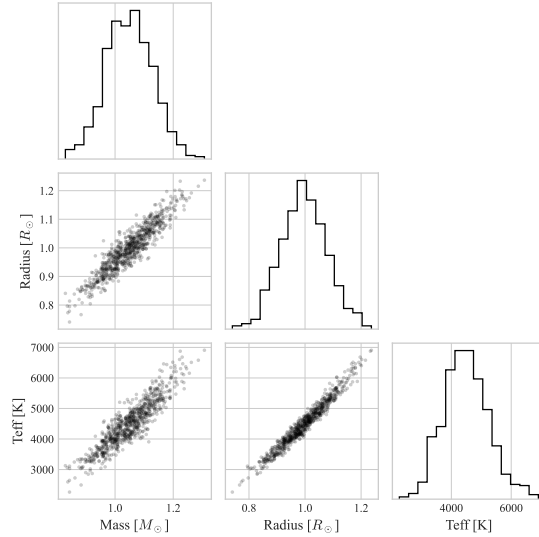


Figure 2: Posterior distributions for the optimal model parameters.

As mapped above in Figure 2, the posterior distributions break down the optimal model parameter correlations.

The derived parameters constrain the planet entirely within the theoretical habitable zone, making it a prime candidate for future spectroscopic atmospheric analysis. The MCMC correlation matrices emphasize that while radius is tightly constrained by the transit geometry, the mass retains a larger variance intrinsically linked to the undefined density prior.

References

J Kepler. A rocky planet in the habitable zone. 2024. doi: 10.1038/nature12345.

D Foreman Mackey. Advanced markov chain monte carlo methods in astrophysics. 2025. doi: 10.1038/nature54321.